

Toward the Evolution of the Desktop GUI: Human-Centered AI as a Paradigm for Rethinking Operating System Interaction

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Abstract. *Innovation in desktop operating system interfaces has stagnated over the past two decades. Despite exponential growth in computational power, tools such as file explorers, terminals, and configuration managers have scarcely evolved since the 1990s. This plateau particularly affects GNU/Linux, whose power remains inaccessible to non-expert users due to command-line interaction models from the 1970s. This paper analyzes desktop interaction stagnation through the lens of Human-Centered AI (HCAI) and presents BOTAS, a voice-based assistant enabling Spanish-speaking users to interact with GNU/Linux via natural language. Evaluation across 200 tasks shows 89% command-generation accuracy and 100% detection of destructive operations.*

1. Introduction

We operate with processors a hundred million times faster than those of the 1980s, yet the fundamental interaction with a desktop operating system has barely changed in twenty years [Jenson 2024]. Scott Jenson—a veteran UX designer who worked on the original Macintosh Finder at Apple and later at Google—argues that “this immobility is not a technical limitation but a capitulation of design” [Jenson 2024]. The industry has moved from the creative transformation that defined early computing—where Xerox PARC concepts were radically reimaged—to shallow cross-platform imitation.

GNU/Linux powers approximately 90% of cloud infrastructure and 100% of the world’s top 500 supercomputers [W3Techs 2023, Top500.org 2023], yet its command-line interface perpetuates a 1970s paradigm [Raymond 2003] demanding cryptic syntax with no feedback about operational risk. Meanwhile, commercial voice assistants handle trivial queries but none addresses deep OS interaction [Luger and Sellen 2016], and tools like ChatGPT suggest commands without executing them or validating safety. This paper argues that Human-Centered AI (HCAI) [Shneiderman 2020, Shneiderman 2022] provides a framework for overcoming this stagnation, and presents BOTAS, a voice-based assistant that materializes HCAI principles for GNU/Linux interaction in Spanish.

This work contributes: (1) an analysis of desktop interface stagnation through the HCAI lens; (2) the presentation of BOTAS, whose evolution from minimum viable prototype to modular system illustrates HCAI in practice; and (3) preliminary evaluation with accuracy metrics, safety detection, and pilot usability feedback.

2. Conceptual Framework and Related Work

2.1. Stagnation of Desktop Interfaces

HCI has traversed periods of genuine innovation—the desktop metaphor, the Web, the touch revolution—that redefined how users form mental models of computing [Norman 1988]. On the desktop, however, the past two decades constitute a functional plateau. Jenson [Jenson 2024] illustrates this with an anecdote: in the early 1980s, while working on the Finder, his team noticed that truncated filenames lost critical information with end-placed ellipses. “I suggested placing the ellipsis in the middle to preserve the file extension—it was implemented in two days and, forty years later, that code is still running” [Jenson 2024]. Today, in the era of large language models, operating systems still employ file-navigation paradigms from three decades ago—a paradox especially acute for GNU/Linux, where the CLI constitutes the primary barrier to a computing platform that dominates professional infrastructure [W3Techs 2023].

2.2. Human-Centered AI

Shneiderman [Shneiderman 2020, Shneiderman 2022] formalized HCAI as a paradigm where AI systems *amplify* human capabilities rather than replace them, built on four principles: **transparency** (users understand system actions), **human control** (users supervise and can reject proposals), **safety** (prevention of harmful actions, even if inadvertently requested—critical where a typo can cause irreversible data loss), and **empowerment** (users progressively learn underlying mechanisms, building autonomy over dependency). Amershi et al. [Amershi et al. 2019] complement this with 18 practical guidelines for human-AI interaction. As AI adoption grows and users discover the possibilities these tools offer, HCAI provides a coherent foundation for rethinking OS interaction: an assistant grounded in these principles can make GNU/Linux accessible while ensuring users learn rather than depend.

2.3. Voice Assistants and the OS Interaction Gap

Advances in speech recognition—particularly Whisper [Radford et al. 2022]—and in LLMs [Brown et al. 2020, OpenAI 2023] have reached practical multilingual maturity. NL2Bash [Lin et al. 2018] explored natural language to Bash translation (36% exact-match accuracy) but without real-time interaction or safety validation. No existing system integrates voice recognition, command execution, safety classification, and educational feedback within an HCAI workflow. This integration gap defines the opportunity this work addresses.

3. Proposal: BOTAS with HCAI Approach

HCAI principles are materialized on GNU/Linux because this is where the intervention yields the greatest impact: a powerful system whose main adoption barrier is interaction, not functionality [Warschauer 2004]. Its openness permits deep integration, and its zero cost eliminates economic barriers—decisive for educational contexts in Latin America.

3.1. Origins and Evolution

BOTAS (*Bot de Operaciones y Tareas Automatizadas del Sistema*) originated during a social service program focused on AI-integrated applications at a Mexican public university. Initially a minimum viable product, it explored whether natural language could

bridge novice users and the command line. Early versions relied entirely on cloud services (Whisper for transcription, GPT-4o for interpretation) with a GTK3 interface and local TTS via eSpeak. Through iterative development, the system incorporated: offline wake-word detection (Vosk), a local intent-recognition module resolving routine requests without LLM calls, automatic distribution detection across six Linux families, a visual state indicator, and structured request logging. The strategy of routing all requests through the LLM was supplemented by local detection, reducing latency and costs; rigid error handling is being redesigned toward clarification dialogues.

3.2. Current Capabilities

BOTAS implements a five-layer pipeline: (1) input via GUI or wake-word; (2) local intent detection through regex patterns; (3) GPT-4o command parsing for complex requests; (4) deterministic safety validation at three risk levels (*safe*, *warning*, *dangerous*); and (5) execution with educational feedback and TTS narration. Supported operations include file management (creation, renaming, moving, search, deletion, editing), system queries (memory, disk, processes, ports, time), process control, networking, permissions, and web search. Every operation includes an explanation of the generated command, fulfilling the empowerment principle. The safety layer requires explicit confirmation for risky operations. Distribution detection (Debian, Fedora, Arch, openSUSE, Alpine, Void) translates package-management commands automatically.

3.3. Scope and Ongoing Developments

The system currently depends on cloud services (Whisper, GPT-4o). We are developing local operation using DeepSeek-R1-0528-Distill-8b—served via local network or on the host machine—with Vosk for speech recognition, eliminating cloud dependency and addressing privacy concerns for institutional environments. An expected accuracy decrease relative to cloud models is acknowledged.

Recognized limitations include difficulty with ambiguous requests, multi-command concatenation in single utterances, technical jargon disambiguation, and absent clarification dialogues. Features under development include improved error recovery, responsive interface layout, and cross-distribution installation tooling. We plan to release BOTAS as open-source software; post-release feedback will inform version 2.0.

4. Preliminary Evaluation

4.1. Functional Testing

BOTAS was evaluated on 200 representative Linux tasks across five categories, specified as Spanish natural-language requests. The dataset was compiled and verified manually by the developer—a constraint we acknowledge, as it may introduce confirmation bias. Table 1 summarizes results.

Error modes: ambiguous phrasing (6%), multi-pipe flag errors (3%), distribution-specific unavailability (2%). Errors were categorized as transcription failures, LLM misinterpretation, or execution-state mismatches. The deterministic safety layer detected 100% of 43 dangerous operations (recursive deletion, privilege escalation, system modifications, permission changes)—a rate guaranteed by regex-based pattern matching independent of model variability. The local intent detector resolved 29% of tasks without LLM calls, at 100% accuracy, saving 500–1500 ms per request.

Table 1. Command generation accuracy by task category

Category	Tasks	Accuracy
File management	45	84%
System information	40	85%
Process control	35	83%
Networking	30	93%
Permissions	25	80%
Total	200	89%

4.2. Pilot Usability Study

Three users with no prior GNU/Linux experience evaluated BOTAS at its MVP stage. While insufficient for statistical generalization—acknowledged prominently—this pilot yielded qualitative insights. Participants found the natural-language execution capability, educational explanations, and safety feedback encouraging; these positive impressions from users who had never used a Linux terminal motivated continued development and this contribution to the HCI community. Constructive feedback identified confusion with verbal syntax, frustration with seemingly excessive safety confirmations, and difficulty with error messages—directly informing the redesign toward clarification dialogues. Latency data: Whisper transcription averages 1–3 s; Vosk wake-word detection responds in under 200 ms.

5. Discussion and Conclusions

Desktop interface stagnation is a design problem, not a technical one: the enabling technologies exist and are mature. HCAI provides a theoretical and practical framework for addressing it, prioritizing user autonomy over opaque automation. From a Latin American perspective, the linguistic barrier excludes users whose documentation is in English, and closed-ecosystem costs burden educational institutions [Warschauer 2004]. Voice interaction in Spanish reduces these barriers without requiring prior CLI expertise, offering guided learning rather than conditioning through repetition.

Future work includes consolidating local operation (DeepSeek-R1 with Vosk), a formal usability study with an expanded sample, cross-distribution validation, and the open-source release.

The HCAI principles of transparency, control, safety, and empowerment—materialized in BOTAS—demonstrate that access barriers to powerful systems can be reduced without sacrificing capabilities. The goal is not to make technology simpler, but to make it more accessible—and in that process, more human.

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